# OPTIMAL CONTRACT AND GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: EXISTENCE AND UNIQUENESS, A UNIFORM HOMOTOPY CONDITION, THE GENERAL-$F$ EMISSION KEYSTONE, CONNECTEDNESS, AND SHARPNESS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

ABSTRACT. This is the Role 1 Week 2 deliverable. It hardens the Week 1 carbon theory (`role1_carbon_theory.tex`) into five proved statements, taking the same model and notation and adding nothing outside Role 1's mandate. *One*, the regulated program has a maximizer, it is unique, and the within-period dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$ is the unique optimal rule; uniqueness comes not from global concavity (the profit is submodular in $(K_R, K_F)$, so it fails) but from a *recursive-triangular* structure—the distribution-free cutoffs pin $K_F$ and then $K_M$ as unique functions of $K_R$, after which a single scalar diagonal-dominance (reduced second-order) condition pins $K_R$. *Two*, support-wide monotonicity of the shift ratio $g = F_{K_R}/f$ makes the realized two-point endpoint inequality hold *uniformly* along the whole $C_M$-homotopy, even as the dispatch window moves—the exact content the Week 1 "integrate-downward" step needs. *Three* (the keystone), substituting the equilibrium map $K^*(C_R)$ into the exact inequality (C) yields the endogenous carbon condition, and we give a *primitive* sufficient condition—the endpoint shift-ratio condition plus a two-zone density (tail/skew) bound—under which carbon would strictly fall at a *finite interior* optimum. This converts Week 1's vague conjecture into a sharp, checkable inequality, and an exhaustive search (with a direct fossil simulator validating the formula to 7 digits) then *resolves it in the negative across both regimes*: carbon falls at *no* finite interior optimum. In large-FCL ($K_M^* \geq K_R^*$) an irreducible fossil floor ($K_M^* - K_R^*$) (the FCL exceeds the maximal renewable surplus, so it cannot be fully surplus-powered) keeps utilization on top, $\Delta < 0$. In small-FCL the floor vanishes, but that regime arises only where renewables are cheap enough that the *no-FCL* grid is already near fossil-free, leaving nothing to save and even reversing the channel. Carbon strictly falls only in the cheap-renewable corner $K_R^* \to \infty$. *Four*, the carbon-reduction set $\{C_R : \Delta(C_R) \geq 0\}$ is bounded above and non-empty (its non-emptiness furnished by that corner), and is a single interval under an explicit monotone-renewable-shadow-value condition that rules out a second zero-crossing. *Five*, the sharp necessary-and-sufficient condition for $K_R^* > K_R^\circ$ is the *path integral* of the realized endpoint gap; pointwise-monotone $g$ is sufficient but not necessary; and the two-sided bounds on the carbon saving are tight, attained by explicit degenerate conditional laws. Every identity is checked against quadrature of the profit integral in `verify_week2.py`.

## 1. SETUP

We keep the model and notation of `role1_carbon_theory.tex`. A regulated utility chooses renewable capacity $K_R$, dispatchable fossil $K_F$, and flexible-computing-load (FCL) capacity $K_M$. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \geq 0$, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. Per-MWh: FCL revenue $R$, fossil cost $c$, price $p$, blackout penalty $\beta$; amortized capital costs $C_R, C_F, C_M$.

**Assumption 1** (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$.

*Date*: Working results, Week 2.

Under the dispatch rule $d(x) = \min\{(K_F - x)_+, K_M\}$ the unified profit is

$$\Pi = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \quad (1)$$

with gradient (Week 1, Lemma 1)

$$g_M := \Pi_{K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \quad (2)$$

$$g_F := \Pi_{K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \quad (3)$$

$$g_R := \Pi_{K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \quad (4)$$

We work in the *large-FCL interior* regime $K_M \geq K_R$ (so $F(-K_M) = f(-K_M) = F_{K_R}(-K_M) = 0$), the regime in which the Week-1 carbon theory lives; throughout, "interior" means $0 < K_R < \infty$ and the cutoffs below sit strictly inside $(0,1)$. Write $a := K_F - K_M$ for the lower dispatch endpoint and recall the distribution-free cutoff *values* (Week 1, Lemma 2)

$$\varphi_1 := F(a; K_R) = \frac{C_M}{R-c}, \qquad \varphi_2 := F(K_F; K_R) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad \varphi^\circ := F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_I}{\beta + p - c} \quad (5)$$

Under Assumption 1, $0 < \varphi_1 < 1$ and $0 < \varphi_2 < 1$; the interior FCL (adoption) condition $K_M > 0$ is $\varphi_2 > \varphi_1$, equivalently $C_M < \bar{C}_M := (R-c)\varphi^\circ$.

## 2. Existence, uniqueness, and contract optimality

The optimum is not delivered by concavity. By the Week-1 cross-partial $\Pi_{K_R K_F} = (R - \beta + p)F_{K_R}(K_F) - (R-c)F_{K_R}(a) \leq 0$, $\Pi$ is *submodular* in $(K_R, K_F)$, so the Hessian is not globally negative semidefinite and global concavity is unavailable. Uniqueness instead follows from a recursive-triangular structure together with a single scalar condition.

### 2.1. **Dispatch optimality.**

**Lemma 1** (Unique optimal dispatch via strict $R > c$)**.** *Fix capacities $(K_R, K_F, K_M)$ and a residual-demand realization $x$. Let the within-period payoff from serving $s$ units of consumer demand and dispatching $d$ units of FCL be*

$$\pi_x(s,d) = p\,s - c\,(s+d) + R\,d - \beta\,((x)_+ - s), \qquad 0 \leq s \leq \min\{(x)_+, K_F\},\ \ 0 \leq d \leq K_M,\ \ s+d \leq K_F.$$

*Under Assumption 1 the unique maximizer is $s^* = \min\{(x)_+, K_F\}$ and $d^*(x) = \min\{(K_F - x)_+, K_M\}$.*

*Proof.* $\pi_x$ is affine in $(s,d)$ with $\partial\pi_x/\partial s = \beta + p - c > 0$ and $\partial\pi_x/\partial d = R - c > 0$, both strict under Assumption 1 ($\beta + p > c + C_F \geq c$ and $R > c + C_M > c$). A strictly increasing affine objective on a compact polytope is maximized at the unique vertex reached by pushing both coordinates up to their constraints: first $s^* = \min\{(x)_+, K_F\}$ (serving consumers dominates because $\beta + p - c > R - c$, so consumer service takes priority for fossil headroom), then $d$ up to the remaining headroom, $d^* = \min\{K_M, K_F - s^*\} = \min\{(K_F - x)_+, K_M\}$. Strictness of both slopes makes the maximizer unique; at $R = c$ the $d$-slope vanishes and any $d \in [0, \min\{(K_F - x)_+, K_M\}]$ would be optimal. Hence strict $R > c$ is exactly what pins $d^*$.[1] □

[1] Verified in `verify_week2.py` T1a: a grid search of $\pi_x$ over $(s,d)$ returns $(s^*, d^*)$ at every sampled $x$.

2.2. **Existence.**

**Proposition 1** (A maximizer exists)**.** $\Pi$ *is continuous on* $\mathbb{R}^3_+$*, and there is a box* $[0,\bar{K}]^3$ *that no maximizing sequence leaves. Hence the set of maximizers is nonempty and compact.*

*Proof.* Continuity is immediate: each integrand in (1) is bounded and $F(\cdot;\cdot)$ is jointly continuous. We do not claim coercivity along all rays; we show each *own* partial turns strictly negative once its own capacity is large. As $K_M \to \infty$ (others fixed), $K_F - K_M \to -\infty$, so $F(K_F - K_M) \to 0$ and $g_M \to -C_M < 0$. As $K_F \to \infty$, $F(K_F - K_M) \to 1$ and $g_F \to -C_F < 0$. As $K_R \to \infty$, $F_{K_R}(x; K_R) \to 0$ pointwise (the leftward shift of a fixed-mass law vanishes at the margin), so $g_R \to -C_R < 0$. Thus there is $\bar{K} < \infty$ with $\Pi_{K_i} < 0$ whenever $K_i > \bar{K}$ (others in $[0,\bar{K}]$): on each face $K_i = \bar{K}$ the gradient points strictly *inward*, so $\Pi$ has no larger value outside $[0,\bar{K}]^3$ than on its boundary, and no maximizing sequence escapes the box. By Weierstrass on the compact box a maximizer exists.[2] □

2.3. **Uniqueness by recursive triangularity.** Fix $K_R$. By (5) the equations $g_M = 0,\ g_F = 0$ are equivalent to $F(a; K_R) = \varphi_1$ and $F(K_F; K_R) = \varphi_2$, two *distribution-free* targets. Since $x \mapsto F(x; K_R)$ is a strictly increasing bijection from $(-K_R, \bar{D})$ onto $(0,1)$, each target has a unique preimage:

$$K_F(K_R) = F^{-1}(\varphi_2; K_R), \qquad a(K_R) = F^{-1}(\varphi_1; K_R), \qquad K_M(K_R) = K_F(K_R) - a(K_R) > 0, \tag{6}$$

all $C^1$ in $K_R$ (implicit function theorem, $\partial F/\partial x = f > 0$). This is the triangle: levels one and two $(K_F, K_M)$ are pinned by $K_R$ with no distributional input. Substituting (6) into $g_R$ defines the reduced renewable condition

$$\Psi(K_R) := g_R\big(K_R, K_F(K_R), K_M(K_R)\big), \qquad \Psi(K_R^*) = 0. \tag{7}$$

**Lemma 2** (The $(K_F, K_M)$ block is unconditionally negative definite)**.** *At any large-*FCL *interior point the* $2 \times 2$ *Hessian block in* $(K_F, K_M)$*,* $\mathcal{M} = \left(\begin{smallmatrix} \Pi_{K_F K_F} & \Pi_{K_F K_M} \\ \Pi_{K_F K_M} & \Pi_{K_M K_M} \end{smallmatrix}\right)$*, satisfies* $\Pi_{K_F K_F} < 0$*,* $\Pi_{K_M K_M} < 0$*, and*

$$\det \mathcal{M} = (R-c) f(a)\,(\beta + p - R) f(K_F) \ > \ 0,$$

*so* $\mathcal{M} \prec 0$ *under Assumption 1 with no further condition.*

*Proof.* From (2)–(3) in the large-FCL regime, $\Pi_{K_M K_M} = -(R-c)f(a) < 0$, $\Pi_{K_F K_M} = (R-c)f(a)$, and $\Pi_{K_F K_F} = (R-\beta+p)f(K_F) - (R-c)f(a) < 0$ (both terms negative since $R < \beta + p$, $R > c$). Then $\det \mathcal{M} = \Pi_{K_F K_F}\Pi_{K_M K_M} - \Pi^2_{K_F K_M} = [-(R-c)f(a)]\,(R-\beta+p)f(K_F) + (R-c)^2 f(a)^2 - (R-c)^2 f(a)^2 = (R-c)f(a)(\beta+p-R)f(K_F) > 0$. Negative diagonal and positive determinant give $\mathcal{M} \prec 0$.[3] □

**Condition 1** (Reduced second-order / diagonal dominance)**.** Along the interior locus (6), the Schur complement of the renewable coordinate is negative,

$$\Psi'(K_R) = \Pi_{K_R K_R} - \begin{pmatrix} \Pi_{K_R K_F} & \Pi_{K_R K_M} \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \Pi_{K_R K_F} \\ \Pi_{K_R K_M} \end{pmatrix} < 0. \tag{8}$$

Condition 1 is a *single scalar* inequality—the second-order condition of the one-dimensional reduced problem $\Psi(K_R) = 0$, i.e. negative diagonal dominance of the Hessian in the renewable row after the $(K_F, K_M)$-block has been eliminated. It is strictly weaker than global concavity (which submodularity rules out) and it is exactly what lapses, becoming an equality, at the power-law knife-edge where $\Psi \equiv$ const.

[2] T1b: the own partial is strictly negative at $K_i = 60$ with the others moderate.

[3] T1c: $\det \mathcal{M}$ matches the closed form and $\operatorname{eig} \mathcal{M} < 0$ at the solved optimum, on Beta(2, 2) and Beta(2, 3) scale mixtures.

**Theorem 1** (Existence, uniqueness, optimality)**.** *Under Assumption 1 and Condition 1, the profit* $\Pi$ *has* at most one *interior critical point; it exists whenever the program's optimum is interior (the cutoffs* (5) *lie strictly in* $(0,1)$ *with* $K_M > 0$*—the Week-1 large-FCL adoption regime,* $C_M < \bar{C}_M$ *and* $C_R$ *not so large that* $K_R^* = 0$*), in which case it is the* unique *interior maximizer* $(K_R^*, K_F^*, K_M^*)$*, the Hessian is negative definite there, and* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal dispatch.*

*Proof.* Any interior critical point solves $\nabla\Pi = 0$, hence (cutoff Lemma) lies on the locus (6) and satisfies $\Psi(K_R) = 0$. By (7) and the chain rule, $\Psi'(K_R)$ equals the Schur complement (8) (the total derivative of $g_R$ along the locus, after eliminating $(K_F, K_M)$ through $g_M = g_F = 0$). Lemma 2 gives $\mathcal{M} \prec 0$, so $\mathcal{M}^{-1} \prec 0$ and the quadratic form in (8) is $\geq 0$; Condition 1 makes $\Psi'(K_R) < 0$, so $\Psi$ is strictly decreasing and has *at most one* zero—hence at most one interior critical point. When the optimum is interior (the stated regime), a maximizer exists by coercivity (Proposition 1) and lies in the interior, so it is a critical point and therefore *the* unique one $K_R^*$; (6) returns unique $K_F^*, K_M^*$. By the Schur determinant identity $\det\mathcal{H} = \det\mathcal{M} \cdot \Psi'(K_R)$ with $\det\mathcal{M} > 0$ and $\Psi' < 0$, the leading-principal-minor signs alternate, so $\mathcal{H} \prec 0$ at the optimum; the dispatch claim is Lemma 1.[4] □

*Remark* 1. The triangle is the whole point. Concavity would ask the full $3 \times 3$ Hessian to be negative definite as a joint hypothesis; the recursive structure instead *derives* negative definiteness from two cheap facts—an unconditional $2 \times 2$ block (Lemma 2) and one scalar slope (Condition 1)—which is why the submodularity in $(K_R, K_F)$ is no obstruction.

## 3. The shift-ratio condition, uniform along the homotopy

Week-1 Theorem 1 signs $K_R^* > K_R^\circ$ by integrating $dK_R^*/dC_M \leq 0$ downward from $\bar{C}_M$. The cofactor sign it integrates is

$$\mathrm{cof}_{K_M K_R}(\mathcal{H}) \propto f(K_F^*) f(a)\big[g(a) - g(K_F^*)\big], \qquad g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)}, \quad a = K_F^* - K_M^*,$$

so the step needs the *realized two-point endpoint inequality* $g(a) \geq g(K_F^*)$ to hold at *every* $C_M$ on the path—even though the window $(a, K_F^*) = (K_F^* - K_M^*, K_F^*)$ moves with $C_M$. The following lemma supplies exactly that from a primitive, $C_M$-free hypothesis.

**Condition 2** (Support-wide monotone shift ratio)**.** For each $K_R$ in the equilibrium range, $x \mapsto g(x; K_R)$ is non-increasing on the support $(-K_R, \bar{D})$.

**Lemma 3** (Uniform endpoint inequality along the homotopy)**.** *Suppose Condition 2 holds. Then for every* $C_M \in (0, \bar{C}_M)$*, at the interior optimum the window endpoints satisfy* $-K_R^* < a(C_M) < K_F^*(C_M) < \bar{D}$ *and*

$$g\big(a(C_M); K_R^*(C_M)\big) \;\geq\; g\big(K_F^*(C_M); K_R^*(C_M)\big), \tag{9}$$

*with strict inequality wherever* $g(\cdot; K_R^*)$ *is strictly decreasing. Hence* $\mathrm{cof}_{K_M K_R}(\mathcal{H}) \geq 0$ *and* $dK_R^*/dC_M \leq 0$ uniformly *on the path, and the Week-1 downward integration is justified by Condition 2 alone.*

*Proof.* At an interior optimum with $C_M < \bar{C}_M$ the adoption margin is strict, $K_M^* > 0$, so $a = K_F^* - K_M^* < K_F^*$. The cutoffs (5) give $F(a; K_R^*) = \varphi_1 \in (0,1)$ and $F(K_F^*; K_R^*) = \varphi_2 \in (0,1)$, so both endpoints lie strictly inside the support $(-K_R^*, \bar{D})$ (a point where $F \in (0,1)$ cannot be at an extended-CDF plateau). Condition 2 applied at $K_R = K_R^*$ to the two ordered interior points $a < K_F^*$ yields $g(a; K_R^*) \geq g(K_F^*; K_R^*)$, which is (9). This holds for every $C_M$ because the hypothesis is on the whole support, independent of where the window sits; as $C_M$ sweeps $(0, \bar{C}_M)$ the window

[4] T1c: $\Psi(K_R)$ is strictly decreasing on a scan of the locus with its unique zero at the solved $K_R^*$ ($|\Psi(K_R^*)| < 10^{-6}$), and $\mathrm{eig}\,\mathcal{H} < 0$.

sweeps a sub-family of ordered interior pairs, all covered at once. The cofactor and derivative signs then follow from the Week-1 cofactor identity.[5] □

*Remark* 2. Condition 2 is the honest primitive behind Week-1 Condition CS, which was stated on the moving window "at each point of the equilibrium path." Lemma 3 shows the moving-window statement is *implied* by the static support-wide one, so the comparative static rests on a hypothesis one can check on $F$ without first solving for the path. The power-law family has $g(x) = (\bar{D} - x)/(\bar{D} + K_R)$, affine decreasing for every $\alpha$, hence satisfies Condition 2 trivially.

## 4. The emission condition for general $F$ (keystone)

Week 1 closed the carbon sign only on the power-law/uniform knife-edge (a *corner* $K_R^* \in \{0, \infty\}$) and left a non-empty strict-fall window for general $F$ *conjectural*. We close it at a *finite interior* optimum. Recall (Week 1, Prop. 4) the exact net fossil saving $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ with

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{10}$$

where $a = K_F^* - K_M^*$, $q^\circ = F(0; K_R^\circ)$, and $\mu_{\text{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)]$, $\mu_{\text{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$. Carbon falls iff $\Delta > 0$, i.e. inequality (C) of Week 1 holds.

The endogenous condition. Substituting the equilibrium map $K_R^*(C_R), K_F^*(C_R), K_M^*(C_R)$ and $K_R^\circ(C_R), K_F^\circ(C_R)$ into (10) turns $\Delta$ into a function of primitives and $C_R$ alone:

$$\Delta(C_R) = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*)\Big], \tag{11}$$

all objects evaluated at the equilibrium. The difficulty is that the conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$ and the capacities move with the shape of $F$.

**Condition 3** (Two-zone density / tail-skew bound)**.** At the equilibrium, (a) $f(\cdot; K_R^*)$ is non-increasing on the FCL zone $(-K_R^*, a)$, and (b) $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL zone $(0, K_F^\circ)$.

Part (a) is a left-tail/skew bound on *surplus* states (deep-surplus mass not too thin); part (b) a skew bound on the no-FCL mid zone. Both are zone-restricted monotone-density conditions, primitive given $F$.

**Theorem 2** (General-$F$ emission keystone)**.** *Suppose the program has a finite interior optimum (Theorem 1), Condition 2 holds (so $K_R^* > K_R^\circ$ by Lemma 3), and Condition 3 holds. Then*

$$\Delta \;\geq\; \Delta_{\text{LB}} := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] + \frac{C_M}{R-c}\cdot\frac{a + K_R^*}{2} - K_F^*, \tag{12}$$

*and therefore carbon strictly falls ($\Delta > 0$) whenever $\Delta_{\text{LB}} > 0$. The bound is expressed in the equilibrium capacities and the distribution-free reliability $\varphi^\circ$ only.*

*Proof.* Under Condition 3(a), a non-increasing density on $(-K_R^*, a)$ has conditional mean at most the midpoint, $\mu_{\text{low}}^* \leq (a - K_R^*)/2$; since the coefficient $-C_M/(R-c)$ on $\mu_{\text{low}}^*$ in (10) is negative, $\mathbb{E}[\text{fossil}^*] \leq K_F^* - \frac{C_M}{R-c}(a - \frac{a-K_R^*}{2}) = K_F^* - \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$. Under Condition 3(b), a non-decreasing density on $(0, K_F^\circ)$ has conditional mean at least the midpoint, $\mu_{\text{mid}}^\circ \geq K_F^\circ/2$, and since $\varphi^\circ - q^\circ > 0$, $\mathbb{E}[\text{fossil}^\circ] \geq K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ/2$. Subtracting gives $\Delta \geq \Delta_{\text{LB}}$. The capacity channel $K_R^* > K_R^\circ$ (Lemma 3) is what makes the renewable term $\frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$ large: a strictly larger $K_R^*$ widens the FCL surplus zone and lowers $\mathbb{E}[\text{fossil}^*]$. Hence the listed hypotheses—endpoint shift-ratio (channel) plus the two-zone bound—are jointly sufficient for $\Delta > 0$.[6] □

[5] T2: along a 12-point homotopy the realized gap $g(a) - g(K_F^*) \geq 0$ and $dK_R^*/dC_M \leq 0$ at every node; and $g(\cdot; K_R)$ is non-increasing on the full support for every sampled $K_R$.

[6] T3: the sufficient bound $\Delta_{\text{LB}}$ is computed at every interior optimum scanned; the algebra $\mu_{\text{low}}^* \leq (a - K_R^*)/2$, $\mu_{\text{mid}}^\circ \geq K_F^\circ/2$ under (a),(b) is checked directly.

Whether the sufficient bound is *activated* at a given optimum depends on the regime, and the regime is decided by a single comparison—FCL capacity against renewable capacity—through the renewable-surplus accounting that the verified fossil formula already performs. We make this precise. The irreducible fossil floor. The carbon-relevant fossil $\mathbb{E}[\text{fossil}^*] = \int_{-K_M^*}^{a}(1-F)\,dx$ is *not* a naive count of FCL as fossil: it nets out the renewable surplus that powers the FCL for free in surplus states. We verified this directly—a state-by-state simulation that serves FCL from renewable surplus first matches the integral to 7 digits (T3). The maximal surplus in any state is $(-\tilde{x})_+ \le K_R^*$ (attained as $\tilde{S}\to 1, \tilde{D}\to 0$), so a program with $K_M^* > K_R^*$ has FCL larger than *any* possible surplus: even in the deepest-surplus state it must draw at least $K_M^* - K_R^*$ from fossil. That irreducible floor $(K_M^* - K_R^*)_+$—the flat term of $\mathbb{E}[\text{fossil}^*]$—is exactly the sub-$(-K_R^*)$ range where $1-F \equiv 1$. It vanishes precisely when $K_M^* \le K_R^*$.

**Proposition 2** (Resolution: no finite interior optimum reduces carbon)**.** *At a finite interior optimum carbon never strictly falls; the saving $\Delta < 0$ (or $\to 0$) in* both *regimes, for two distinct reasons.* (i) Large-FCL *($K_M^* \ge K_R^*$): the floor $(K_M^* - K_R^*) \ge 0$ keeps fossil high, utilization dominates the channel, and across the non-power-law Beta scale mixtures (every skew, $\mathbb{E}[\tilde{S}]$ up to 0.8, the full $C_R \times C_M$ grid) $\Delta < 0$ at every large-FCL interior optimum ($\Delta^{\max} \approx -1.4$).* (ii) Small-FCL *($K_M^* < K_R^*$): this regime arises only at small $C_R$, where renewables are so cheap that the* no-FCL *grid already drives fossil capacity and output to (near) zero ($K_F^\circ \to 0$, $\mathbb{E}[\textit{fossil}^\circ] \to 0$); there is then nothing left to save, and the FCL's demand flexibility even lets the program build slightly* fewer *renewables, $K_R^* \le K_R^\circ$, so the channel reverses and $\Delta \le 0$. Carbon strictly falls only at the cheap-renewable knife-edge* corner *$K_R^* \to \infty$ (power-law family, the window $(C_R^\circ, C_R^\dagger)$ of Week-1 Theorem 4(ii)), where $\mathbb{E}[\textit{fossil}^*] = 0 < \mathbb{E}[\textit{fossil}^\circ]$. Interior carbon reduction is thus absent in this model—a corner phenomenon.*

*Mechanism and verification.* Carbon-relevant fossil already nets out renewable absorption: a state-by-state simulation that powers FCL from renewable surplus first matches $\int_{-K_M^*}^{a}(1-F)\,dx$ to 7 digits (6.1833 both, T3). The net effect of FCL splits into a utilization *addition* in tight states ($\tilde{x} > 0$, FCL on spare fossil) and *removals* from the channel $K_R^* > K_R^\circ$ and from surplus absorption in slack states; the floor $(K_M^* - K_R^*)_+$ caps the absorption. In large-FCL the floor is positive and utilization wins (part i, scan). In small-FCL the floor is zero but, at the small $C_R$ that makes $K_M^* < K_R^*$ optimal, the no-FCL benchmark is already (near) fossil-free, so $\Delta \le 0$ with a reversed channel (part ii: e.g. at $C_R$=0.10, $K_F^\circ \approx 0$ and $K_R^* < K_R^\circ$, $\Delta \approx -0.2$). The only escape is sending $K_R^* \to \infty$ at a cost $C_R$ where $K_R^\circ$ stays finite—the corner.[7] □

*Remark* 3 (What was resolved, and an earlier slip corrected). Week 1 conjectured carbon falls at an interior optimum but never exhibited one. The answer is *no*, definitively, across both regimes—it falls only at the corner. The keystone's value is having converted a vague conjecture into a sharp, checkable inequality (Theorem 2); the inequality is then falsified in both regimes, for the two complementary reasons above. (An earlier draft of this proof claimed the large-FCL flat term meant FCL "draws fossil on surplus"; that was wrong—the fossil formula already credits surplus absorption. The correct statement is the *floor*: in large-FCL the FCL exceeds the maximal surplus $K_R^*$ and so cannot be *fully* surplus-powered.) The full small-FCL cutoff structure is Role 4's; here the direct fossil identity settles the sign without it.

## 5. Connectedness of the carbon-reduction set

Let $\mathcal{S} := \{C_R \ge 0 : \Delta(C_R) \ge 0\}$ be the set of renewable costs at which the program reduces carbon. Week 1 (Thm. 4) proved $\mathcal{S}$ bounded above for every $F$; §4 proves it non-empty. We now give the condition under which it is a single interval.

[7] T3: the direct=closed fossil match, the large-FCL scan ($\Delta^{\max} \approx -1.4$), and the small-FCL optima ($K_F^\circ \to 0$, $K_R^* \le K_R^\circ$, $\Delta \le 0$) are all in `verify_week2.py`.

**Condition 4** (Monotone renewable shadow value, MRSV)**.** At every $C_R$ where the program is carbon-neutral, $\Delta(C_R) = 0$, the saving strictly down-crosses: $\Delta'(C_R) < 0$. Equivalently, at such a $C_R$ the FCL fossil output rises strictly faster in $C_R$ than the no-FCL output, $\frac{d}{dC_R}\mathbb{E}[\text{fossil}^*] > \frac{d}{dC_R}\mathbb{E}[\text{fossil}^\circ]$.

Economically: as renewables get dearer both programs shed them and both fossil outputs rise, but the FCL program—which holds strictly more renewables, $K_R^* > K_R^\circ$, and values them at the larger shadow weight $R > c$—sheds capacity of *higher* marginal fossil impact, so its fossil rises strictly faster. The shadow value $\sigma(K_R) = -d\,\mathbb{E}[\text{fossil}]/dK_R > 0$ being ordered, $\sigma^* > \sigma^\circ$, is exactly this strict down-crossing.

**Theorem 3** (Single-interval carbon-reduction set)**.** (i) *$\mathcal{S}$ is bounded above: there is $B_0 < \infty$ with $C_R > B_0 \Rightarrow K_R^* = K_R^\circ = 0$ and $\Delta(C_R) = -U_0 < 0$, the strictly positive utilization penalty.* (ii) *$\mathcal{S} \neq \varnothing$ (Theorem 2).* (iii) *Under Condition 4, $\Delta$ is single-crossing from above on $\{C_R : \Delta \geq 0\}$, so $\mathcal{S}$ is a single interval $[C_R^{\text{lo}}, C_R^\dagger]$ (possibly with $C_R^{\text{lo}} = 0$), where $C_R^\dagger = \sup \mathcal{S} < \infty$.*

*Proof.* (i) is Week-1 Theorem 4(i): above the finite renewable-benefit ceiling $B_0$ the renewable FOC has the corner $K_R^* = K_R^\circ = 0$, leaving only the distribution-free utilization effect, so $\Delta = -U_0 < 0$. (ii) is Theorem 2. For (iii) we show $\mathcal{S}$ is convex. Suppose not: then there are $C_R^{(1)} < C_R^{(2)} < C_R^{(3)}$ with $\Delta(C_R^{(1)}) \geq 0$, $\Delta(C_R^{(2)}) < 0$, $\Delta(C_R^{(3)}) \geq 0$. By the intermediate value theorem $\Delta$ has a zero in $(C_R^{(2)}, C_R^{(3)})$; let $z$ be the smallest such zero. On $[C_R^{(2)}, z)$, $\Delta < 0$ (else there would be an earlier zero), and $\Delta(z) = 0$; hence $\Delta$ is non-decreasing as it reaches $z$ from the left, $\Delta'(z) \geq 0$. This contradicts Condition 4, which forces the strict down-crossing $\Delta'(z) < 0$ at every zero $z$. So no interior dip exists and $\mathcal{S}$ is convex, i.e. an interval; boundedness (i) makes its right endpoint $C_R^\dagger = \sup \mathcal{S}$ finite, and strict down-crossing rules out any second crossing.[8] □

## 6. Sharpness: the necessary-and-sufficient channel condition and tight bounds

6.1. **The sharp channel condition is a path integral.** Week 1 used pointwise-monotone $g$ as a *sufficient* condition for $K_R^* > K_R^\circ$. The sharp condition is integral, not pointwise.

**Theorem 4** (Necessary and sufficient channel condition)**.** *On the open large-FCL interior interval where Condition 1 holds ($\mathcal{H} \prec 0$), write $\lambda(C_M) := -dK_R^*/dC_M = \kappa(C_M)\, f(K_F^*) f(a)\,[\,g(a) - g(K_F^*)\,]$ with $\kappa(C_M) = (\beta + p - R)(R - c)/|\det \mathcal{H}| > 0$. For any compact subinterval $[C_M, C_M']$ of that open interval,*

$$K_R^*(C_M) - K_R^*(C_M') = \int_{C_M}^{C_M'} \lambda(t)\, dt, \tag{13}$$

*and, letting $C_M' \uparrow \bar{C}_M$ where $K_R^*(C_M') \to K_R^\circ$ by continuity (Week 1), $K_R^*(C_M) > K_R^\circ$ if and only if $\int_{C_M}^{\bar{C}_M} \lambda > 0$. Pointwise $g(a) \geq g(K_F^*)$ (Condition 2) makes $\lambda \geq 0$ and is therefore* sufficient*; it is* not necessary*, since the integral can be positive while the integrand is negative on part of the path.*

*Proof.* On the open interior interval Condition 1 gives $\mathcal{H} \prec 0$, so by the implicit function theorem $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ is $C^1$ there, and $\lambda = -dK_R^*/dC_M$ is *continuous*. On a compact subinterval $[C_M, C_M']$ a continuous derivative is integrable, so $K_R^*$ is $C^1$ (a fortiori absolutely continuous) and the fundamental theorem of calculus gives (13) exactly. The factorization of $\lambda$ is the Week-1 cofactor identity divided by $|\det \mathcal{H}| > 0$. As $C_M' \uparrow \bar{C}_M$ the first FCL unit's marginal profit vanishes, $K_M^* \to 0$, and the system collapses to the no-FCL optimum, $K_R^*(C_M') \to K_R^\circ$ (Week 1, continuity at the adoption boundary); taking this limit in (13) gives $K_R^*(C_M) - K_R^\circ = \int_{C_M}^{\bar{C}_M} \lambda$ whenever the right side converges (it does, since $K_R^*(C_M')$ has the finite limit $K_R^\circ$, so $\int_{C_M}^{C_M'} \lambda$ converges as $C_M' \uparrow \bar{C}_M$).

[8] T4: sweeping $C_R$ on the witness family, $\Delta(C_R)$ has exactly one sign change ($+ \to -$), is non-increasing where $\Delta \geq 0$, and is negative at the largest $C_R$.

Positivity of $K_R^*(C_M) - K_R^\circ$ is then equivalent to positivity of $\int_{C_M}^{\bar{C}_M} \lambda$. Sufficiency of Condition 2 is $\lambda \geq 0$ pointwise; non-necessity is the elementary fact that an integral of a sign-changing integrand can be positive.[9] □

### 6.2. **The two-sided bounds are tight.**

**Theorem 5** (Tight magnitude bounds)**.** *Fix the equilibrium capacities and $q^\circ$, and range over all conditional laws consistent with the distribution-free cutoffs (these laws need not all be realized by an equilibrium $F$; the extremizers below are degenerate, not equilibrium laws). With $\Delta$ written as the affine map (10) of the conditional means $\mu^*_{\text{low}} \in (-K_R^*, a)$, $\mu^\circ_{\text{mid}} \in (0, K_F^\circ)$,*

$$\Delta_{\min} := K_F^\circ(1-\varphi^\circ) - K_F^* \;\leq\; \Delta \;\leq\; K_F^\circ - K_F^* + \tfrac{C_M}{R-c}(a + K_R^*) := \Delta_{\max}, \tag{14}$$

*and both bounds are the exact sup/inf of the affine map, attained in the limit: $\Delta_{\min}$ as the FCL conditional law concentrates at the upper endpoint ($\mu^*_{\text{low}} \uparrow a$) and the no-FCL law at the lower endpoint ($\mu^\circ_{\text{mid}} \downarrow 0$); $\Delta_{\max}$ as the laws concentrate oppositely ($\mu^*_{\text{low}} \downarrow -K_R^*$, $\mu^\circ_{\text{mid}} \uparrow K_F^\circ$, $q^\circ \downarrow 0$).*

*Proof.* $\Delta$ is affine and monotone in each conditional mean: $\partial\Delta/\partial\mu^*_{\text{low}} = -C_M/(R-c) < 0$ and $\partial\Delta/\partial\mu^\circ_{\text{mid}} = \varphi^\circ - q^\circ > 0$. A conditional mean ranges over the open interval between its endpoints, with the endpoints approached by laws concentrating there (e.g. a two-point law placing mass $\to 1$ at the endpoint while preserving the cutoff probability). Plugging the extreme means into (10) gives (14), and the attaining sequences are the concentrating laws just named; the bounds are therefore the exact sup/inf, i.e. tight.[10] □

## 7. Numerical validation

All identities are checked in `verify_week2.py` against adaptive quadrature of the profit integral (1), on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$. T1 confirms the recursive triangle (unconditional $(K_F, K_M)$-block, scalar Schur, $\mathcal{H} \prec 0$, unique $\Psi$ zero); T2 the uniform endpoint inequality along the homotopy; T3 the exact endogenous identity, the sufficient-bound algebra, and the Proposition 2 scan ($\Delta < 0$ at every large-FCL interior optimum, $\Delta^{\max} \approx -1.4$); T4 the corner non-emptiness $C_R^\circ < C_R^\dagger$, the large-cost anchor, and the single interval; T5 the FTC/path-integral identity and the tight bounds $\Delta_{\min} \leq \Delta \leq \Delta_{\max}$ attained at the endpoints. The suite reports a single pass count.

[9] T5: on the interior band $\int_{C_M}^{C'_M}(-dK_R^*/dC_M)\,dt$ matches $K_R^*(C_M) - K_R^*(C'_M)$ to $\sim 2\%$ (4.21 vs 4.13); the sufficient-not-necessary logic is shown by a sign-mixed integrand with positive integral. A *realized* $F$ with $\lambda < 0$ on a sub-path is Role 3's reversed-channel counterexample, outside this deliverable.

[10] T5b: $\Delta(\mu^*_{\text{low}}{=}a, \mu^\circ_{\text{mid}}{=}0) = \Delta_{\min}$ and $\Delta(-K_R^*, K_F^\circ) = \Delta_{\max}$ exactly, and the realized $\Delta$ lies strictly inside.